

Quantum feature maps have been proposed as a route to machine-learning advantage, yet two separate questions are of never observed, 0/48 configurations), and under noise: depolarization preserves the central relations, while amplitude damp

I. INTRODUCTION

The promise of quantum machine learning rests on a single question: does embedding classical data into an exponential Two lines of recent work have questioned the optimistic picture. On one side, quantum kernels can suffer *exponential c* This has motivated a practical ambition: before paying the cost of quantum computation or training, predict *whether a* Related work approaches this ambition from several directions. Kübler et al. [7] analyzed the inductive bias of quantum Yet the two questions such criteria are asked to answer are logically different. *Classical simulability* is a property of the In this work we ask: *what do spectral criteria actually predict?* We adopt a task-resolved numerical protocol, using only

A. Contributions

Assumptions. All results below are numerical and obtained with exact state-vector simulation of $n \leq 5$ qubit circuits
Main results.

1. Spectral criteria reliably predict *classical simulability* as measured by RFF and Nyström approximation error; this is a clear
2. Spectral criteria do *not* predict quantum advantage globally. Across six tasks, the correlation between effective rank and ad
3. The dominant predictor of advantage is the *saturation of classical baselines* on the task: the best classical test accuracy cor
4. The frequency scale of the feature map is the decisive design knob for spectral spread and simulability ($\rho = +1.000$, $p = 1.4$
5. The central relations are robust to scaling, real data, and noise: the simulability relation strengthens at 8–12 qubits ($\rho = +$

II. DEFINITIONS AND BACKGROUND

A. Quantum feature maps and fidelity kernels

A quantum feature map $\phi: \mathbb{R}^d \rightarrow \mathcal{H}$ maps a classical feature vector x to a normalized state $\phi(x) = U(x)|0 \cdots 0\rangle$, where

B. Spectral criteria

Let $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$ be the eigenvalues of K , and $p_i = \lambda_i / \sum_j \lambda_j$. We use the following criteria, all computable in O

- **Effective rank** (participation ratio): $r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, ranging from 1 (perfectly concentrated) to $\text{rank}(K)$.
- **Spectral entropy**: $H = -\sum_i p_i \log_2 p_i$, with maximum $\log_2 N$.
- **Concentration index**: $c = 1 - r_{\text{eff}}/N$.
- **Gap ratio**: $\lambda_1 / (\lambda_1 + \lambda_2)$.
- **Top- k energy**: $\sum_{i \leq k} \lambda_i / \sum_i \lambda_i$ ($k = 5$).

C. Classical simulability metrics

Given K and the original data matrix X , we measure approximability by two classical families. *Random Fourier featur*

D. Quantum advantage and classical baselines

For a task with training/test split, define

where acc_Q is the test accuracy of a kernel-SVM using the quantum kernel, and the maximum is over a strong classical base

E. Tasks

Tasks have controlled difficulty and structure: (i) *harmonic* (sum of fixed-frequency sinusoids), (ii) *periodic-coupled* (co

III. METHODS

A. Simulator and feature maps



